CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

3.3 Linear Momentum & its Conservation

Syllabus Section: 3. Dynamics **Paper Type:** Structured Questions (Paper 2) **Total:** 12 Questions (148 Marks)

Question 1 Easy [10 marks]

Part (a) [2 marks]

State the principle of conservation of linear momentum.

Mark Scheme (2 marks):

The principle of conservation of linear momentum states that:

- The sum / total momentum (of a system) is constant **OR** the sum / total momentum before equals the sum / total momentum after *[1 mark]*
- If no (resultant) external force acts **OR** for an isolated system *[1 mark]*

Many students write "in a closed system" as the condition. It sounds similar, but it does not say that no external force acts, so it will not earn the second mark. Finish your statement with "provided no resultant external force acts" and you have the condition the mark needs.

Part (b) [2 marks]

A pellet of mass $20\ \text{g}$ is fired from an air rifle with a momentum of $1.2\ \text{kg m s}^{-1}$.

Calculate the velocity of the pellet just after the rifle is fired.

Mark Scheme (2 marks):

To calculate the velocity of the pellet just after the rifle is fired:

- List the known quantities:
  - Mass of the pellet, $m = 20\ \text{g} = 0.020\ \text{kg}$
  - Momentum of the pellet, $p = 1.2\ \text{kg m s}^{-1}$
- Write down the equation for momentum and rearrange it for *v*:

$$p = mv$$

$$v = \frac{p}{m}$$

- Substitute the known values:

$$v = \frac{1.2}{0.020}$$ *[1 mark]*

- Calculate the answer:

$$v = 60\ \text{m s}^{-1}$$ *[1 mark]*

Convert the mass to kilograms before you divide. Leaving it in grams gives 0.06 m s^{-1}, which is far too slow for a pellet leaving a rifle, so a quick sense-check of your answer will catch this slip.

Part (c) [2 marks]

Use the principle of conservation of momentum to state:

(i) the total momentum of the air rifle and the pellet

[1]

(ii) the recoil momentum of the air rifle.

[1]

Mark Scheme (2 marks):

(i) The total momentum of the air rifle and the pellet is:

- Zero *[1 mark]*

(ii) The recoil momentum of the air rifle is:

- $1.2\ \text{kg m s}^{-1}$ (in the opposite direction to the pellet) *[1 mark]*

$-1.2\ \text{kg m s}^{-1}$ is also accepted. The direction does not have to be stated, because the word "recoil" already gives it

Momentum is a vector, so its direction matters as well as its size. The rifle and the pellet start at rest, so their momenta after firing must cancel out, and that is why the rifle moves backwards. Here the word "recoil" already tells you the direction, but in other questions make the direction clear with a minus sign or by saying which way the object moves.

Part (d) **[4 marks]**

The pellet hits a target. It becomes embedded and takes $0.0025\ \text{s}$ to come to rest.

(i) State the relation between force and momentum.

[1]

(ii) Calculate the average force needed to stop the pellet.

[3]

Mark Scheme (4 marks):

(i) The relation between force and momentum is:

- (Resultant) force equals the rate of change of momentum **OR** $F = \frac{\Delta p}{\Delta t}$ *[1 mark]*

"Force is proportional to the rate of change of momentum" is also accepted

(ii) To calculate the average force needed to stop the pellet:

- List the known quantities:
  - Momentum of the pellet before hitting the target, $p = 1.2\ \text{kg m s}^{-1}$
  - Momentum of the pellet when it comes to rest = 0

- Time taken, $\Delta t = 0.0025$ s

- Calculate the change in momentum:

$$\Delta p = 1.2 - 0 = 1.2 \text{ kg m s}^{-1} \textit{ [1 mark]}$$

- Write down the equation and substitute the known values:

$$F = \frac{\Delta p}{\Delta t} = \frac{1.2}{0.0025} \textit{ [1 mark]}$$

- Calculate the answer:

$$F = 480 \text{ N} \textit{ [1 mark]}$$

1 mark for the correct final answer of 480 N; -480 N is also accepted, where the minus sign shows that the force acts against the pellet's motion

Question 2 Easy [11 marks]

Part (a) [3 marks]

A collision can be described as being elastic or inelastic when there are no external forces acting on the collision.

In Table 1.1, place a tick (✓) next to the quantities that are conserved in each type of collision

Table 1.1

Quantity	Elastic Collision	Inelastic Collision
Momentum		
Total Energy		
Kinetic Energy		

Mark Scheme (3 marks):

A table that would score 3 marks looks like this:

Quantity	Elastic Collision	Inelastic Collision
Momentum	✓	✓
Total Energy	✓	✓
Kinetic Energy	✓	

- Momentum conserved in **both** elastic and inelastic collisions; *[1 mark]*
- Total energy conserved in **both** elastic and inelastic collisions; *[1 mark]*
- Kinetic energy conserved in elastic collisions **only**; *[1 mark]*

[Total: 3 marks]

*In a closed system (i.e., no matter or mass moves in or out), momentum is **always** conserved during a collision, provided **no external force acts** on the system. The law of conservation of energy can never be violated, therefore the **total** energy before and after the collision must be conserved. Kinetic energy is only conserved in*

elastic collisions. During inelastic collisions some of the kinetic energy gets converted into other forms such as thermal energy.

Part (b) [3 marks]

A trolley **X** with mass of 400 g is moving on a track with a velocity of 1.2 m s^{-1} towards a stationary trolley **Y** with mass of 800 g as shown in Fig. 1.1.

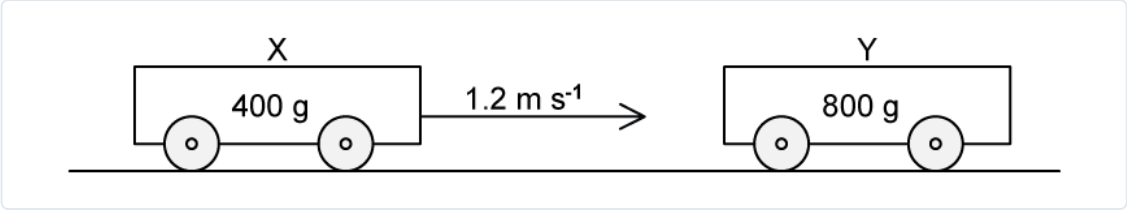


Fig. 1.1

(i) Calculate the initial momentum of trolley **X**.

[2]

(ii) Trolley **X** collides with trolley **Y** and subsequently, they stick together.

State the combined momentum of the trolleys after the collision.

[1]

Mark Scheme (3 marks):

List the known quantities:

- Mass of **X** = 400 g = 0.4 kg
- Velocity of **X** = 1.2 m s^{-1}

(i) Calculate the initial momentum of trolley **X**:

- Momentum: $p = mv$ *[1 mark]*
- Momentum before: p = 0.4 × 1.2
- Momentum before: p = 0.48 kg m s^{-1} *[1 mark]*

(ii) The combined momentum after the collision is:

- Momentum before the collision = momentum after the collision
- Momentum after: p = 0.48 kg m s^{-1} *[1 mark]*

[Total: 3 marks]

Part (c) [4 marks]

(i) Calculate the velocity of the two trucks after the collision.

[3]

(ii) State the direction of the velocity of the trolleys after the collision.

[1]

Mark Scheme (4 marks):

(i) Calculate the velocity of the two trucks after the collision:

List the known quantities:

- Mass of X = 0.4 kg
- Mass of Y = 0.8 kg
- Momentum of the combined trucks, p = 0.48 kg m s^{-1}

Determine the new mass when the trucks are stuck together:

- Total mass after collision = (0.4 + 0.8) = 1.2 kg *[1 mark]*

Rearrange the momentum equation and substitute in the values:

- $p = mv \quad \Rightarrow \quad v = \frac{p}{m}$
- Velocity after: $v = \frac{0.48}{1.2}$ *[1 mark]*
- Velocity after: v = 0.4 m s^{-1} *[1 mark]*

(ii) The direction of the final velocity will be:

- The same direction / to the right; *[1 mark]*

[Total: 4 marks]

*Since velocity is a vector, and so is momentum, the positive or negative sign tells you its direction. In this scenario, the initial direction of trolley **X** has been taken as positive. Since the final momentum is positive, this implies the trolleys are moving in the same direction as trolley **X** at the beginning.*

Part (d) [1 mark]

Explain why the collision cannot be described as elastic.

Mark Scheme (1 marks):

The collision is not elastic because:

- The final velocity / kinetic energy is <u>less</u> after the collision

  OR

- The relative speed of separation is <u>less than</u> the relative speed of approach; *[1 mark]*

[Total: 1 mark]

If you're not sure how to calculate the relative speeds, it is simply:

- *The relative speed of approach:* $u_X - u_Y = 1.2 - 0 = 1.2 \text{ m s}^{-1}$
- *The relative speed of separation:* $v_{XY} = 0.4 \text{ m s}^{-1}$

It's useful to note that when two objects stick together after a collision, they can ***never*** *be elastic*

In fact when two objects stick together, it is said to be "perfectly" inelastic, because, in this scenario, the maximum amount of kinetic energy is dissipated.

You can think of this as the remaining energy wasted as heat or sound. If there was a collision of two cars where one drives into a stationary one, you would hear the sound of the collision.

Question 3 Easy [9 marks]

Part (a) [2 marks]

State the difference between an elastic and an inelastic collision.

Mark Scheme (2 marks):

The difference between an elastic and an inelastic collision is:

- In an elastic collision, the total kinetic energy is conserved **OR** relative speed of approach = relative speed of separation; *[1 mark]*
- In an inelastic collision, kinetic energy is not conserved / is lost **OR** relative speed of approach > relative speed of separation; *[1 mark]*

[Total: 2 marks]

Part (b) [2 marks]

A ball collides with a wall at a velocity of 35 m s^{-1} and rebounds with velocity *v*. The ball has a mass of 0.025 kg.

The motion of the ball is shown in Fig. 1.1.

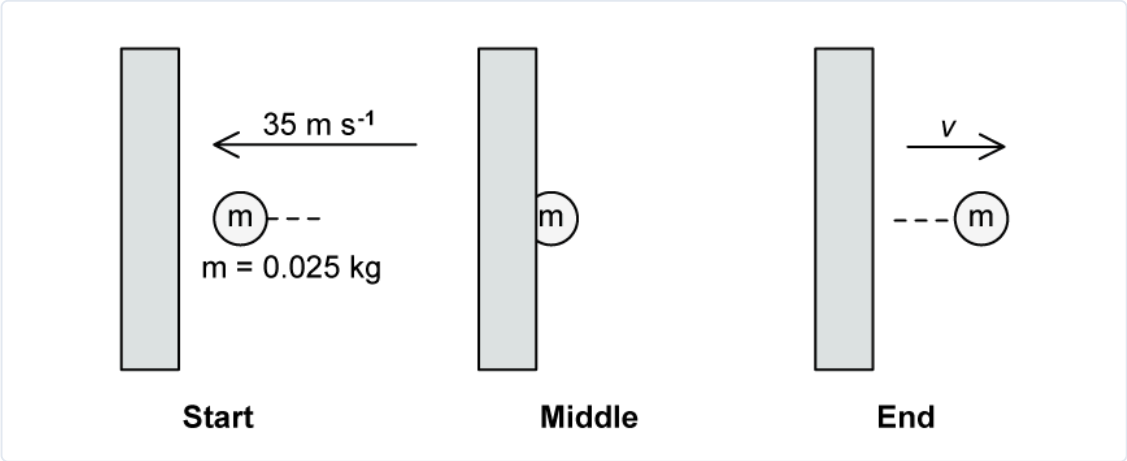


Fig. 1.1

Calculate the momentum of the ball at the start, before it hits the wall.

Mark Scheme (2 marks):

Calculate the momentum of the ball at the start, before it hits the wall:

List the known quantities:

- Mass, *m* = 0.025 kg
- Velocity, *v* = 35 m s^{-1}

State the equation for momentum and calculate:

- Momentum: $p = mv$ *[1 mark]*
- Momentum: $p = 0.025 \times 35$
- Momentum: $p = 0.875 = 0.88$ kg m s^{-1} *[1 mark]*

[Total: 2 marks]

Part (c) **[3 marks]**

The collision is perfectly elastic.

Determine

(i) the rebound velocity of the ball

[1]

(ii) the magnitude of the change in momentum of the ball.

[2]

Mark Scheme (3 marks):

List the known quantities:

- Mass, m = 0.025 kg
- Initial momentum = 0.875 kg m s^{-1}
- Initial velocity = 35 m s^{-1}

(i) Determine the rebound velocity of the ball:

- For a perfectly elastic collision: relative speed of approach = relative speed of separation
- Hence, rebound (final) velocity: $v = -35$ m s^{-1} *[1 mark]*

(ii) Determine the change in momentum of the ball:

- Change in momentum: Momentum after − momentum before
- Change in momentum: Δp = -mv - mv = -2mv
- Change in momentum: $\Delta p = (0.025 \times -35) - (0.025 \times 35)$ **OR** 2×0.875 *[1 mark]*
- Magnitude of the change in momentum: $\Delta p = 1.75 = 1.8$ kg m s^{-1} *[1 mark]*

[Total: 3 marks]

The question demonstrates the importance of direction. Since momentum is a vector quantity, the direction influences the sign on the value.

*The **magnitude** of the change in momentum means don't worry about the signs, the final answer should just be the **size**. However, the negative or positive signs are important to get to this stage!*

*If momentum was a scalar, you would say the 'change in momentum' was zero, as the magnitude of the momentum before and after didn't change. However, the fact that the momentum acts in the opposite direction afterwards means the magnitude of the total momentum is **twice** that of the momenta before and after the collision.*

This very fact has crucial implications in the real world, such as in gas laws, where it can be used to explain why gas pressure exists

Part (d) **[2 marks]**

The ball is in contact with the wall for 0.1 s.

Calculate the force exerted on the ball by the wall.

Mark Scheme (2 marks):

Calculate the force exerted on the ball by the wall

List the known quantities:

- Change in momentum, Δp = 1.75 kg m s^{-1}
- Change in time, Δt = 0.1 s

Use the equation for force and momentum:

- $F = \frac{\Delta p}{\Delta t}$ *[1 mark]*
- $F = \frac{1.75}{0.1}$
- Force: F = 17.5 = 18 N *[1 mark]*

[Total: 2 marks]

Question 4 Medium [13 marks]

Part (a) [3 marks]

A ball X of mass 140 g travelling at constant velocity *v* collides head-on with a stationary ball Y of mass 620 g, as shown in Fig. 1.1.

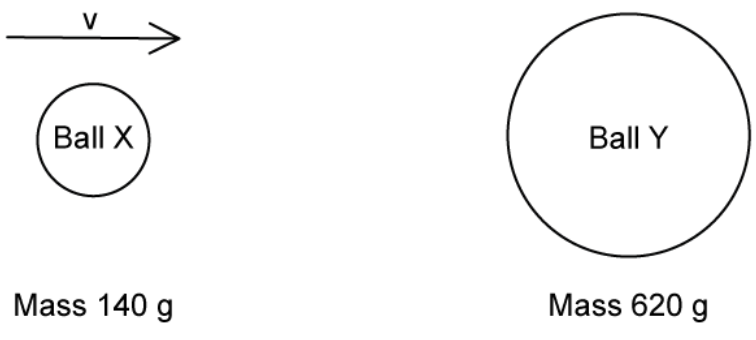


Fig. 1.1

After the collision, ball X comes to a stop. Ball Y moves off in the same initial direction as X with a velocity of 0.8 m s^{-1}.

Determine the velocity *v* of ball X before the collision.

Mark Scheme (3 marks):

Determine the velocity *v* of ball X before the collision:

List the known quantities:

- Mass of X = 140 g
- Mass of Y = 620 g
- Velocity of X before the collision = *v*
- Velocity of X after the collision = 0
- Velocity of Y before the collision = 0
- Velocity of Y after the collision = 0.8 m s^{-1}

Use conservation of momentum to write an expression for *v*:

- Momentum before collision: $m_X v_X + m_Y v_Y = 140v + 0$
- Momentum after collision: $m_X v_X + m_Y v_Y = 0 + 620 \times 0.8$
- Conservation of momentum: Momentum before collision = Momentum after collision
- Conservation of momentum: $140v = 620 \times 0.8$ *[1 mark]*
- Rearrange: $v = \frac{620 \times 0.8}{140}$ *[1 mark]*

- v = 3.543 = 3.5 m s^{-1} *[1 mark]*

[Total: 3 marks]

Part (b) **[4 marks]**

(i) Show that the collision in **(a)** is inelastic.

[3]

(ii) Explain why the collision in **(a)** is inelastic.

[1]

Mark Scheme (4 marks):

(i) Show that the collision in **(a)** is inelastic:

List the known quantities:

- Mass of X = 140 g = 0.14 kg
- Mass of Y = 620 g = 0.62 kg
- Velocity of X before the collision = 3.5 m s^{-1}
- Velocity of X after the collision = 0
- Velocity of Y before the collision = 0
- Velocity of Y after the collision = 0.8 m s^{-1}

Determine the kinetic energy before and after the collision:

- KE before = $\frac{1}{2}m_X {v_X}^2 + \frac{1}{2}m_Y {v_Y}^2 = \frac{1}{2} \times 0.14 \times 3.5^2 + 0$
- KE before = 0.8575 = 0.86 J *[1 mark]*
- KE after = $\frac{1}{2}m_X {v_X}^2 + \frac{1}{2}m_Y {v_Y}^2 = 0 + \frac{1}{2} \times 0.62 \times 0.8^2$
- KE after = 0.1984 = 0.20 J *[1 mark]*

Write a concluding sentence:

- The kinetic energy after the collision is <u>less</u> than the kinetic energy before the collision, hence it is an inelastic collision; *[1 mark]*

(ii) The collision is inelastic because:

- Some of X's (kinetic) energy has been transferred to other forms of energy (such as heat and sound); *[1 mark]*

[Total: 4 marks]

Part (c) **[3 marks]**

The balls are replaced by an aluminium sphere of mass 2.7 kg and a steel sphere of 7.9 kg as shown in Fig. 1.2.

The aluminium sphere has an initial velocity of 10.0 m s^{-1} when it collides with the stationary steel sphere. Immediately after the collision, the velocity of the steel sphere is 5.1 m s^{-1}.

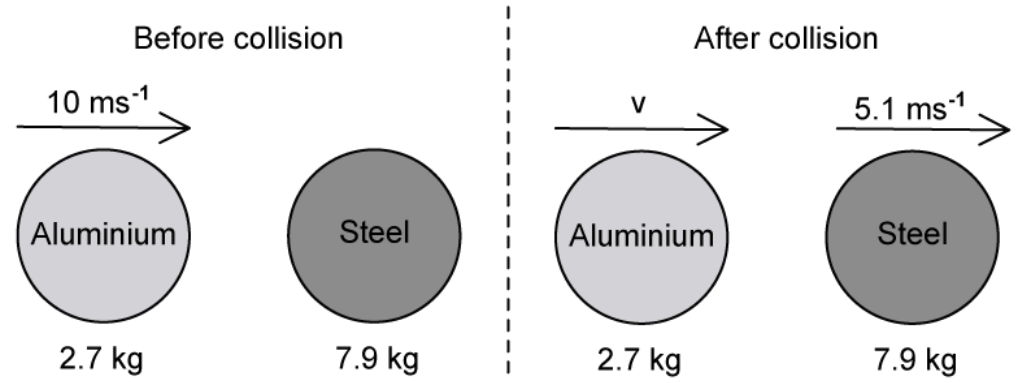


Fig. 1.2

Calculate the velocity *v* of the aluminium sphere immediately after the collision.

Mark Scheme (3 marks):

Calculate the velocity of the aluminium sphere immediately after the collision:

List the known quantities:

- Mass of aluminium sphere, m_{Al} = 2.7 kg
- Mass of steel sphere, m_S = 7.9 kg
- Initial velocity of aluminium sphere, u_{Al} = 10.0 m s^{-1}
- Initial velocity of steel sphere, u_S = 0
- Final velocity of aluminium sphere = v
- Final velocity of steel sphere, v_S = 5.1 m s^{-1}

Use conservation of momentum to write an expression for *v*:

- Momentum before collision: $m_{Al}u_{Al} + m_S u_S = 2.7 \times 10 + 0$

  AND

- Momentum after collision: $m_{Al}v_{Al} + m_S v_S = 2.7 \times v + 7.9 \times 5.1$ *[1 mark]*
- Conservation of momentum: Momentum before collision = Momentum after collision
- $2.7 \times 10 = 2.7v + 7.9 \times 5.1 \Rightarrow v = \frac{27 - 7.9 \times 5.1}{2.7}$ *[1 mark]*
- v = −4.922 = −4.9 m s^{-1} *[1 mark]*

[Total: 3 marks]

- *The negative sign is necessary for the final answer*
    - *This tells us that the aluminium sphere **changes** direction*
- *Remember to show your working - the first mark is for **substitution**, not just stating the equation!*

Part (d) [3 marks]

Verify that the collision in **(c)** is elastic.

Mark Scheme (3 marks):

Verify that the collision in **(c)** is elastic:

List the known quantities:

- Mass of aluminium sphere, m_{Al} = 2.7 kg
- Mass of steel sphere, m_S = 7.9 kg
- Initial velocity of aluminium sphere, u_{Al} = 10.0 m s^{-1}
- Initial velocity of steel sphere, u_S = 0
- Final velocity of aluminium sphere, v_{Al} = 4.9 m s^{-1}
- Final velocity of steel sphere, v_S = 5.1 m s^{-1}

Determine the kinetic energy before and after the collision:

- KE before = $\frac{1}{2}m_{Al}{u_{Al}}^2 + \frac{1}{2}m_S{u_S}^2 = \frac{1}{2} \times 2.7 \times 10^2 + 0$
- KE before = 135 J *[1 mark]*
- KE after = $\frac{1}{2}m_{Al}{v_{Al}}^2 + \frac{1}{2}m_S{v_S}^2 = \frac{1}{2} \times 2.7 \times 4.9^2 + \frac{1}{2} \times 7.9 \times 5.1^2$
- KE after = 135.15 = 135 J *[1 mark]*

Write a concluding sentence:

- The kinetic energy after the collision is the same as the kinetic energy before the collision, hence it is an elastic collision; *[1 mark]*

[Total: 3 marks]

Question 5 Medium [13 marks]

Part (a) [3 marks]

Two identical blocks A and B each of mass 400 g are travelling towards each other along a straight line through their centre, as shown in Fig. 1.1.



Fig. 1.1

Both blocks are moving at a speed of 0.36 m s^{-1} relative to the surface.

As a result of the collision, the blocks reverse their direction of motion and travel at the same speed as each other. During the collision, 33% of the kinetic energy of the blocks is transferred to the surroundings as thermal energy.

(i) State the total momentum of the blocks before the collision.

[1]

(ii) State and explain whether the collision is elastic or inelastic.

[2]

Mark Scheme (3 marks):

(i) The total momentum of the blocks before the collision is:

- Total momentum = *mu* + *mu* = (400 × 0.36) + (400 × −0.36)
- Total momentum = 0 *[1 mark]*

(ii) The collision is:

- Inelastic; *[1 mark]*

Because:

- 33% of the kinetic energy is lost as heat **OR** in elastic collisions, 100% of kinetic energy is conserved; *[1 mark]*

[Total: 3 marks]

Most scenarios where two objects collide and then move in opposite directions are commonly elastic. However, there is information given in the question that suggests that this isn't the case for this scenario.

Always read the full question carefully, and don't just assume one or the other!

Part (b) [3 marks]

Calculate the final speed of the blocks.

Mark Scheme (3 marks):

Calculate the final speed of the blocks:

List the known quantities:

- Energy lost = 33% = 0.33 $\Rightarrow \quad E_{k2} = 0.67 \times E_{k1}$
- Initial speed of each block, u = 0.36 m s^{-1}
- Final speed of each block = v

Write an expression for the ratio of the initial and final kinetic energies:

- Kinetic energy: $E_k = \frac{1}{2}mv^2 \quad \Rightarrow \quad E_k \propto v^2$
- Initial kinetic energy: $E_{k1} \propto u^2$
- Final kinetic energy: $E_{k2} \propto v^2$
- $\frac{E_{k1}}{E_{k2}} = \frac{u^2}{v^2}$ *[1 mark]*

Determine the final speed of the blocks, v:

- $\frac{E_{k1}}{0.67 \times E_{k1}} = \frac{0.36^2}{v^2} \quad \Rightarrow \quad v^2 = 0.36^2 \times 0.67$
- $v = 0.36 \times \sqrt{0.67}$ *[1 mark]*
- v = 0.2947 = 0.29 m s^{-1} *[1 mark]*

[Total: 3 marks]

Part (c) [3 marks]

The duration of the collision between the blocks is 750 ms.

Determine the average force exerted by one block on the other.

Mark Scheme (3 marks):

Determine the average force exerted by one block on the other:

List the known quantities:

- Initial speed of both blocks, u = 0.36 m s^{-1}

- Final speed of both blocks, v = 0.29 m s^{-1} (from **(b)**)
- Mass of each block, m = 400 g = 0.4 kg
- Time during the collision, Δt = 750 ms = 750 × 10^{-3} s

Calculate the change in velocity, Δv

- $\Delta v = v - u$ = 0.36 – (–0.29) = 0.65 m s^{-1} *[1 mark]*

Calculate the average force, F:

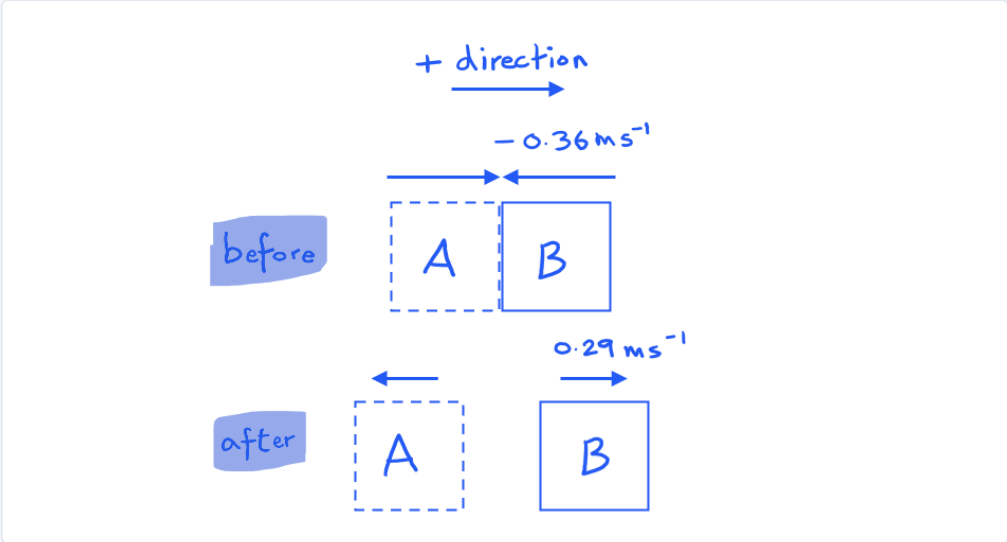
- Momentum: $p = mv$
- Force: $F = \frac{\Delta p}{\Delta t} = \frac{m\Delta v}{\Delta t}$
- $F = \frac{0.4 \times 0.65}{750 \times 10^{-3}}$ *[1 mark]*
- F = 0.35 N *[1 mark]*

[Total: 3 marks]

A common mistake in this question is calculating the change in velocity Δv as 0.36 – 0.29 = 0.07 m s^{-1}

*This is incorrect as it is not treating velocity as a **vector** which has both magnitude and direction*

Make sure to consider just one of the blocks, and be clear on which direction you are taking as the positive direction:



The Δv equation is always final velocity v – initial velocity u. Whether you take the right or left as positive, the magnitude of the force as the final answer will still be the same.

Part (d) **[4 marks]**

The same blocks are investigated with different initial speeds. Block A is launched towards B at a speed of 0.6 m s^{-1} and B is launched towards A at a speed of 0.4 m s^{-1}.

Once the blocks collide, they stick together and move as one after the collision.

Determine the speed of the combined blocks after the collision and state the direction.

Mark Scheme (4 marks):

Determine the speed and direction of the combined blocks after the collision:

List the known quantities:

- Initial velocity of block A, v_A = 0.6 m s^{-1}
- Initial velocity of block B, v_B = −0.4 m s^{-1}
- Mass of block A, m_A = 400 g = 0.4 kg
- Mass of block B, m_B = 400 g = 0.4 kg

Use conservation of linear momentum:

- Conservation of momentum: Total momentum before the collision = Total momentum after the collision
- Conservation of momentum: $m_A v_A + m_B v_B = m_A + m_B v$ *[1 mark]*
- Rearrange: $v = \frac{m_A v_A + m_B v_B}{m_A + m_B} \Rightarrow v = \frac{m v_A + v_B}{2m}$
- $v = \frac{0.4 (0.6 - 0.4)}{2 \times 0.4}$ *[1 mark]*
- v = 0.1 m s^{-1} *[1 mark]*

State the final speed and direction:

- Speed after the collision = 0.1 m s^{-1}
- Direction = same as initial direction of A; *[1 mark]*

[Total: 4 marks]

Question 6 Medium [13 marks]

Part (a) [4 marks]

A fast-moving asteroid B collides with a slower-moving asteroid A as shown in Fig. 1.1. Asteroid B becomes embedded within asteroid A as shown.

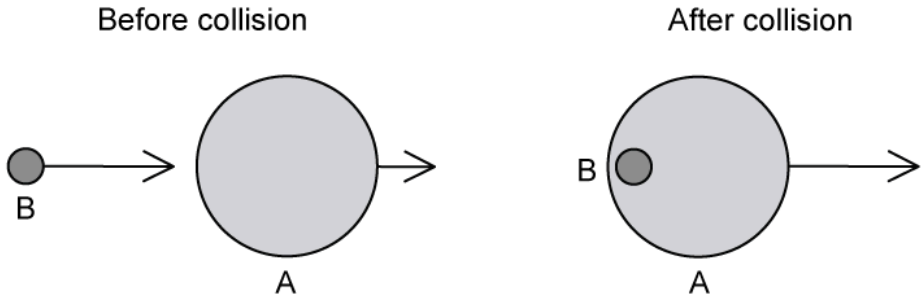


Fig. 1.1

Before the collision, asteroid A had a velocity of 3.61 km s^{-1} and a momentum of 2.00×10^{17} kg m s^{-1}.

(i) State the principle of conservation of linear momentum.

[2]

(ii) Show that the mass of asteroid A was about 5.5×10^{13} kg.

[2]

Mark Scheme (4 marks):

(i) The principle of conservation of linear momentum states that:

- The momentum of a system remains constant **OR** total momentum before (a collision) = total momentum after (a collision); *[1 mark]*
- Provided that there are no external / resultant forces acting; *[1 mark]*

(ii) Show that the mass of asteroid A was about 5.5×10^{13} kg:

List the known quantities:

- Velocity of A, m_A = 3.61 km s^{-1} = 3.61×10^3 m s^{-1}
- Momentum of A, p_A = 2.00×10^{17} kg m s^{-1}

Determine the mass of A:

- $p = mv$ *[1 mark]*
- $m = \frac{p}{v}$
- $m_A = \frac{2.00 \times 10^{17}}{3.61 \times 10^3}$ = 5.54×10^{13} kg *[1 mark]*

[Total: 4 marks]

It's very common for students to miss the second marking point from their definitions of conservation of momentum - make sure you learn both parts!

Remember to use the unrounded calculated value for the mass of asteroid A

You should always work with more sig figs than you need during your calculation and then round at the end to avoid a cumulative rounding error

Part (b) **[3 marks]**

Calculate the combined velocity, in km s^{-1}, of the asteroids after the collision.

mass of asteroid B = 6.40 × 10^{12} kg

velocity of asteroid B before the collision = 14.0 km s^{-1}

Mark Scheme (3 marks):

Calculate the combined velocity of the asteroids after the collision:

- Initial velocity of asteroid A, u_A = 3.61 × 10^3 m s^{-1}
- Momentum of asteroid A, p_A = 2.00 × 10^{17} kg m s^{-1}
- Mass of asteroid B, m_B = 6.40 × 10^{12} kg
- Initial velocity of asteroid B, u_B = 14.0 km s^{-1} = 14.0 × 10^3 m s^{-1}

Apply conservation of momentum:

- After the collision, asteroid B is absorbed by asteroid A, so the asteroids have a combined mass of $m_A + m_B$
- $p_{before} = p_{after}$
- $p_A + m_B u_B = m_A + m_B \; v$ *[1 mark]*

Rearrange and calculate the combined velocity v:

- $v = \frac{p_A + m_B u_B}{m_{AB}}$
- $v = \frac{2.00 \times 10^{17} + 6.40 \times 10^{12} \times 14.0 \times 10^3}{5.54 \times 10^{13} + 6.40 \times 10^{12}}$ *[1 mark]*
- v = 4686 = 4.69 km s^{-1} (3 s.f.) *[1 mark]*

[Total: 3 marks]

Part (c) **[3 marks]**

A short time later, the combined asteroid AB collides with a stationary asteroid C. Immediately following the collision, the asteroids are deflected by the angles shown in Fig. 1.2.

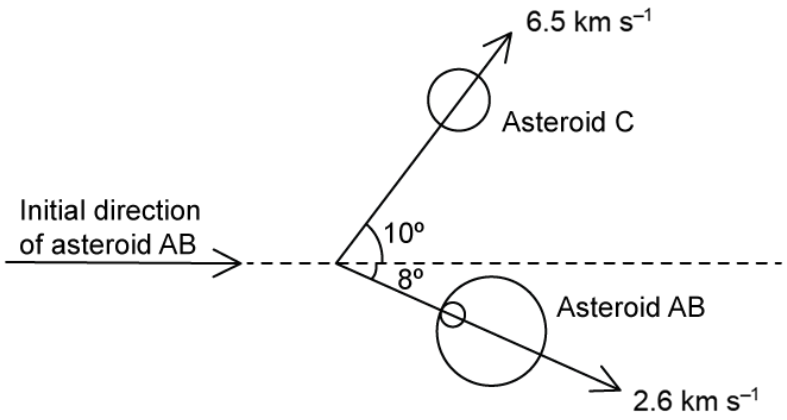


Fig. 1.2

Show that the mass of asteroid C is about 2.0 × 10^{13} kg.

Mark Scheme (3 marks):

Show that the mass of asteroid C is about 2.0 × 10^{13} kg:

List the known quantities:

- Mass of asteroid AB, m_{AB} = (6.4 × 10^{12}) + (5.54 × 10^{13}) = 6.18 × 10^{13} kg
- Mass of asteroid C = m_C
- Initial velocity of asteroid AB, u_{AB} = 4.69 km s^{-1} (from (b))
- Final velocity of asteroid AB, v_{AB} = 2.6 km s^{-1}
- Final velocity of asteroid C, v_C = 6.5 km s^{-1}
- Angle between AB and the horizontal, θ_{AB} = 8°
- Angle between C and the horizontal, θ_C = 10°

Calculate the initial momentum of the system:

- $p_i = m_{AB} v_{AB}$
- $p_i = 6.18 \times 10^{13} \times 4.69 \times 10^{3}$
- p_i = 2.896 × 10^{17} kg m s^{-1} *[1 mark]*

Calculate the final momentum of the system:

- $p_f = m_{AB} v_{AB} \cos \theta_{AB} + m_C v_C \cos \theta_C$
- $p_f = 6.18 \times 10^{13} \times 2.6 \times 10^{3} \times \cos 8° + m_C \times 6.5 \times 10^{3} \times \cos 10°$
- $p_f = 1.607 \times 10^{17} \cos 8° + m_C 6.5 \times 10^{3} \cos 10°$ *[1 mark]*

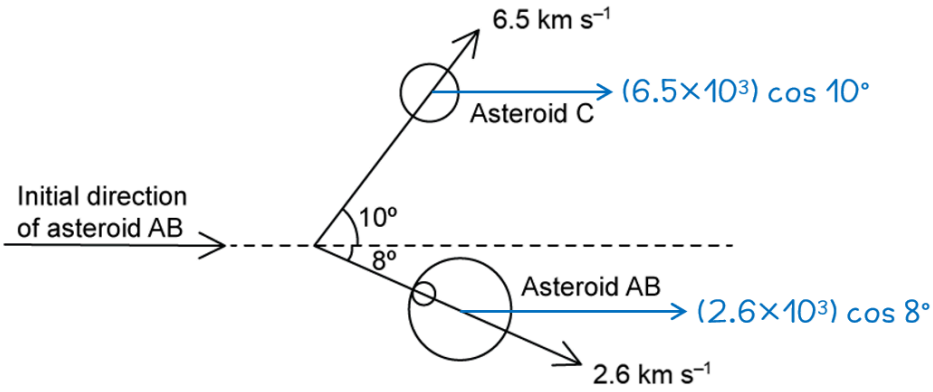
Use the conservation of momentum to calculate the mass of asteroid C:

- Total initial momentum = total final momentum, or $p_i = p_f$
- $2.896 \times 10^{17} = 1.607 \times 10^{17} \cos 8° + m_C 6.5 \times 10^3 \cos 10°$
- $m_C = \frac{2.896 \times 10^{17} - 1.607 \times 10^{17} \cos 8°}{6.5 \times 10^3 \cos 10°}$ = 2.04 × 10^{13} kg *[1 mark]*

[Total: 3 marks]

Make sure you get lots of practice with momentum conservation in 2D scenarios; the extra step of resolving vector components can trip up many students!

*We only need to consider the **horizontal components** of the velocities of the asteroids:*



Part (d) [3 marks]

Determine if the collision in **(c)** is elastic or inelastic.

Mark Scheme (3 marks):

Determine if the collision in **(c)** is elastic or inelastic:

List the known quantities:

- Mass of asteroid AB, m_{AB} = (6.4 × 10^{12}) + (5.54 × 10^{13}) = 6.18 × 10^{13} kg
- Mass of asteroid C, m_C = 2.04 × 10^{13} kg (from (c))
- Initial velocity of asteroid AB, u_{AB} = 4.69 km s^{-1} (from (b))
- Final velocity of asteroid AB, v_{AB} = 2.6 km s^{-1}
- Final velocity of asteroid C, v_C = 6.5 km s^{-1}

Calculate the initial kinetic energy of the system:

- $KE_i = \frac{1}{2} m_{AB} u_{AB}^2$

- $\mathrm{KE}_i = \frac{1}{2} \times 6.18 \times 10^{13} \times 4.69 \times 10^{3}{}^2$
- KE_i = 6.8 × 10^{20} J *[1 mark]*

Calculate the final kinetic energy of the system:

- $\mathrm{KE}_f = \frac{1}{2} m_{AB} v_{AB}{}^2 + \frac{1}{2} m_C v_C{}^2$
- $\mathrm{KE}_f = \frac{1}{2} \times 6.18 \times 10^{13} \times 2.6 \times 10^{3}{}^2 + \frac{1}{2} \times 2.04 \times 10^{13} \times 6.5 \times 10^{3}{}^2$
- KE_f = 6.4 × 10^{20} J *[1 mark]*

Conclusion:

- An inelastic collision is one where kinetic energy is not conserved
- Since $\mathrm{KE}_i > \mathrm{KE}_f$, energy has dissipated, so the collision is inelastic; *[1 mark]*

[Total: 3 marks]

Question 7 Medium [10 marks]

Part (a) [2 marks]

Ball X has a mass of 50 g and is supported by a long string, as shown in Fig. 1.1.

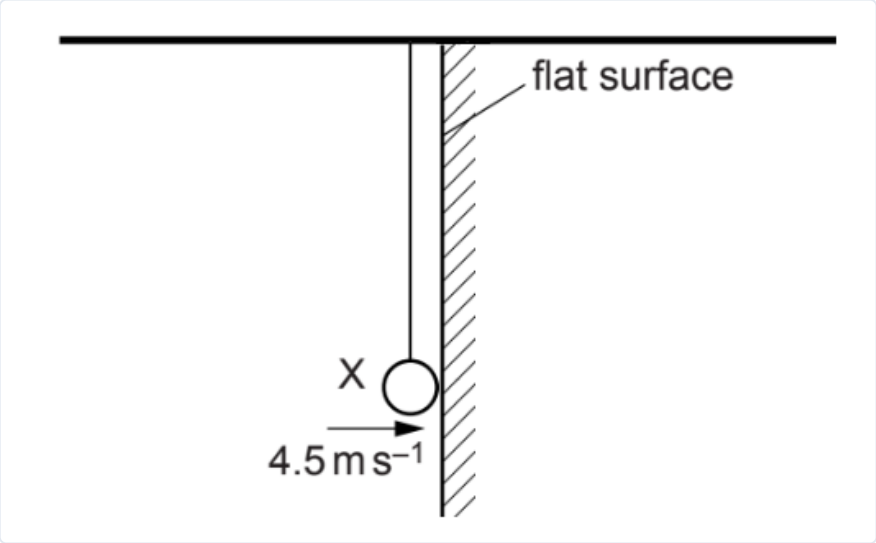


Fig. 1.1

The ball is pulled back and released. The ball collides normally with a flat surface at a velocity of 4.5 m s^{-1} and rebounds elastically.

The positive direction is horizontal and to the right.

Determine the total change in momentum of ball X.

Mark Scheme (2 marks):

Determine the total change in momentum of the ball:

In an elastic collision:

- KE before = KE after
- Therefore, initial velocity = final velocity (= 4.5 m s^{-1}) *[1 mark]*

The change in momentum is:

- Momentum before = mv
- Momentum after = -mv
- Change in momentum: Δp = -mv - mv = - 2mv
- Change in momentum: Δp = -2 × (50 × 10^{-3}) × 4.5 = -0.45 kg m s^{-1} *[1 mark]*

[Total: 2 marks]

Part (b) [3 marks]

Ball X is hung next to ball Y which is supported by a string of the same length, as shown in Fig. 1.2.

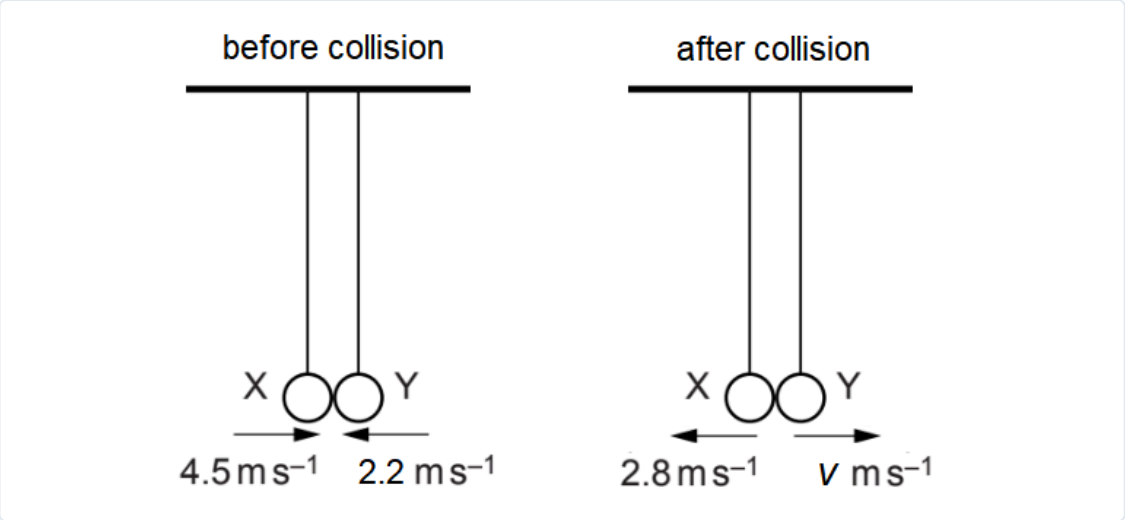


Fig. 1.2

Ball Y has a mass of 125 g.

The balls are each pulled back and pushed towards each other. When the balls collide, the strings are vertical. The balls rebound in opposite directions.

The velocities of X and Y just before and just after the collision are shown in Fig. 1.2.

Use the conservation of linear momentum to determine the rebound velocity v of Y.

Mark Scheme (3 marks):

Use the conservation of linear momentum to determine the rebound velocity of Y.

List the known quantities:

- Mass of X, m_X = 50 g
- Mass of Y, m_Y = 125 g
- Initial velocity of X, u_X = 4.5 m s^{-1}
- Final velocity of X, v_X = −2.8 m s^{-1}
- Initial velocity of Y, u_Y = −2.2 m s^{-1}
- Final velocity of Y, $v_Y = v$

The momenta before and after the collision:

- Momentum before = $m_X u_X + m_Y u_Y$
- Momentum before = (50 × 4.5) + (125 × −2.2) *[1 mark]*
- Momentum after = $m_X v_X + m_Y v_Y$

- Momentum after = (50 × −2.8) + (125 × v) *[1 mark]*

Conservation of linear momentum:

- Momentum before = momentum after: $m_X u_X + m_Y u_Y = m_X v_X + m_Y v_Y$
- (50 × 4.5) + (125 × −2.2) = (50 × −2.8) + (125 × v)
- 225 − 275 = −140 + 125v $\Rightarrow$ $125v = 90$
- Rebound velocity of Y: $v = \frac{90}{125} = 0.72 = 0.7$ m s^{-1} *[1 mark]*

[Total: 3 marks]

In this question, you do not need to convert the mass of the balls to kg. Both balls have their mass in grams and when divided in g or kg the ratio will be the same.

Part (c) **[2 marks]**

Deduce whether the collision in **(b)** is elastic or inelastic.

Mark Scheme (2 marks):

Deduce whether the collision is elastic or inelastic

List the known quantities:

- Initial velocity of X, u_X = 4.5 m s^{-1}
- Final velocity of X, v_X = −2.8 m s^{-1}
- Initial velocity of Y, u_Y = −2.2 m s^{-1}
- Final velocity of Y, v_Y = 0.7 m s^{-1} (from (b))

An elastic collision is...

- A collision in which kinetic energy is conserved **or** relative speed of approach = relative speed of separation

Write a conclusion:

- Calculation showing that total initial KE ≠ the total final KE **or** relative speed of approach ≠ relative speed of separation; *[1 mark]*

  e.g. the relative speeds of approach and separation are:

  - Relative speed of approach: $u_X - u_Y$ = 4.5 − (−2.2) = 6.7 m s^{-1}
  - Relative speed of separation: $v_Y - v_X$ = 0.7 − (−2.8) = 3.5 m s^{-1}
  - The relative speed of approach (6.7 m s^{-1}) ≠ relative speed of separation (3.5 m s^{-1})
- Therefore, the collision is <u>not elastic</u> **or** the collision is <u>inelastic</u>; *[1 mark]*

[Total: 2 marks]

Part (d) **[3 marks]**

Use Newton's laws to explain why the magnitude of the change in momentum of each ball is the same.

Mark Scheme (3 marks):

Use Newton's laws to explain why the magnitude of the change in momentum of each ball is the same

Newton's third law states...

- When two objects interact, they exert equal and opposite forces on each other
- Therefore, the force on X is equal (in magnitude) and opposite (in direction) to the force on Y; *[1 mark]*

Newton's second law states...

- The resultant force on an object is equal to the rate of change of momentum (i.e. $\Delta p/\Delta t$) of the object, and it acts in the same direction, or $F = \Delta p/\Delta t$
- Therefore, the force (on X & Y) is equal to / proportional to the rate of change of momentum; *[1 mark]*

The change in momentum of X & Y is the same because...

- The time of collision is the same for both balls (hence change in momentum is the same); *[1 mark]*

[Total: 3 marks]

Question 8 Medium [10 marks]

Part (a) [2 marks]

A shell of mass 30 g is fired from the barrel of a rifle of mass 1.9 kg. Immediately after being fired, the bullet has a momentum of 36 kg m s^{-1}.

State the total momentum of the rifle and the bullet before the rifle is fired and give a reason for your answer.

Mark Scheme (2 marks):

The total momentum of the rifle and the bullet before being fired is:

- Zero / 0 kg m s^{-1} *[1 mark]*

Because:

- (Before the rifle is fired both) rifle and bullet are stationary / velocity = 0; *[1 mark]*

[Total: 2 marks]

Part (b) [2 marks]

Calculate the velocity of the shell just after the rifle is fired.

Mark Scheme (2 marks):

Calculate the velocity of the shell just after the rifle is fired

List the known quantities:

- Mass of shell, m_s = 30 g = 0.03 kg
- Momentum of shell after being fired, p = 36 kg m s^{-1}

Calculate the velocity of the shell using the momentum equation:

- Momentum: $p = mv$
- $v = \frac{p}{m} = \frac{36}{0.03}$ *[1 mark]*
- $v = 1200$ m s^{-1} *[1 mark]*

[Total: 2 marks]

*For the velocity to be in m s^{-1}, the momentum must be in kg m s^{-1} and the mass in **kg** - unit conversions are very common, don't let these catch you out!*

Part (c) [3 marks]

(i) Using the principle of conservation of momentum, and your answer to part (a), determine the total momentum of the rifle and the shell.

[1]

(ii) Calculate the recoil momentum of the rifle.

[2]

Mark Scheme (3 marks):

(i) The total momentum of the rifle and the shell after the rifle is fired is:

- Total momentum before the rifle is fired = total momentum after the rifle is fired **OR** $p_{initial} + p_{final} = 0$
- Total momentum = 0 kg m s^{-1} *[1 mark]*

(ii) Calculate the recoil momentum of the rifle:

List the known quantities:

- Total final momentum of the rifle and shell, $p_f = 0$
- Momentum of the shell after being fired, p_s = 36 kg ms^{-1} (from **(b)**)

Decide on the positive direction:

- Since p_s = 36 kg m s^{-1} then the motion of the shell gives the positive direction
- Therefore the recoil will have a negative value **OR** $p_s = - p_r$

Calculate:

- Therefore $p_s = - p_r = 36$ *[1 mark]*
- p_r = −36 kg m s^{-1} *[1 mark]*

[Total: 3 marks]

*You may have chosen the recoil of the rifle to represent the **positive** direction. In this case, your values should be the same as ours, but the negative and positive signs will be reversed.*

Part (d) **[3 marks]**

The shell has a momentum of 9.8 kg m s^{-1} just before it hits a target.

It takes 4.5 ms for the bullet to be stopped by the target.

Calculate the average force needed to stop the bullet.

Mark Scheme (3 marks):

Calculate the average force needed to stop the bullet

List the known quantities:

- Momentum of shell before hitting target, p_s = 9.8 kg m s^{-1}
- Time taken for shell to stop, Δt = 4.5 ms = 4.5 × 10^{-3} s

Determine the change in momentum Δp as the bullet is stopped by target:

- The shell comes to rest, therefore final momentum = 0
- Change in momentum: Δp = 9.8 kg m s^{-1} *[1 mark]*

Calculate the average force *F* needed to stop the bullet:

- Force & momentum: $F = \frac{\Delta p}{\Delta t}$
- Average force: $F = \frac{9.8}{4.5 \times 10^{-3}}$ *[1 mark]*
- Average force: $F = 2177.8 = 2200 \text{ N } 2 \text{ s.f.}$ *[1 mark]*

[Total: 3 marks]

Calculations like this one are usually given what examiners call 'error carried forward' (you will see it written 'ecf' in mark schemes).

This means that, for example, if you forgot to convert ms to s (using 4.5 s rather than 0.0045) then you could still get the other three marks, as long as your final answer was 2200 N.

However, if you did get this then your 'sanity check' at the end should have seen you looking at this number and realising it must be wrong, since it suggests that being shot by a rifle feels similar to having someone pat you lightly on the back!

Question 9 Medium [6 marks]

Part (a) [2 marks]

State the principle of conservation of momentum.

Mark Scheme (2 marks):

The principle of conservation of momentum states:

- Total momentum before = total momentum after **OR** the total momentum (of a system) is constant *[1 mark]*
- Provided no (resultant) external forces act on the system **OR** for a closed system *[1 mark]*

Part (b) [4 marks]

Two spheres, A and B, are moving in a straight line on a smooth horizontal surface. Sphere A has a mass of 0.20 kg and a velocity of 5.0 m s^{-1}. Sphere B has a mass of 0.30 kg and a velocity of 2.0 m s^{-1}. Both spheres move in the same direction, as shown in Fig. 3.1.

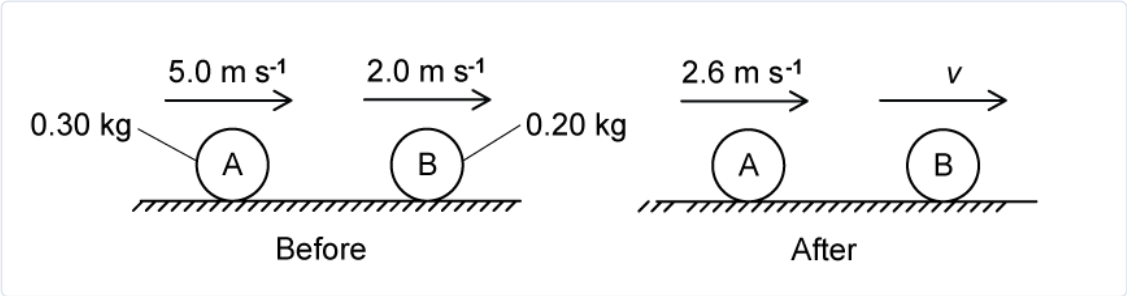


Fig. 3.1

Sphere A collides with sphere B. After the collision, sphere A continues to move in the same direction with a velocity of 2.6 m s^{-1}.

(i) Calculate the velocity v of sphere B after the collision.

[2]

(ii) By calculation of kinetic energies, state and explain whether the collision of the spheres is inelastic or perfectly elastic.

[2]

Mark Scheme (4 marks):

(i) To calculate the velocity v of sphere B after the collision:

- List the known quantities:
  - Mass of A, m_A = 0.30 kg

- Mass of B, m_B = 0.20 kg
- Initial velocity of A, u_A = 5.0 m s^{-1}
- Initial velocity of B, u_B = 2.0 m s^{-1}
- Final velocity of A, v_A = 2.6 m s^{-1}
- Final velocity of B, $v_B = v$

- Determine the momentum of the spheres before and after the collision

$$p = mv$$

$$\text{Total momentum before} = m_A u_A + m_B u_B$$

$$\text{Total momentum before} = 0.30 \times 5.0 + 0.20 \times 2.0$$

$$\text{Total momentum after} = m_A v_A + m_B v_B$$

$$\text{Total momentum after} = 0.30 \times 2.6 + 0.20 \times v \textit{ [1 mark]}$$

- Apply conservation of momentum and calculate v:

$$\text{Total momentum before} = \text{total momentum after}$$

$$0.30 \times 5.0 + 0.20 \times 2.0 = 0.30 \times 2.6 + 0.20 \times v$$

$$v = \frac{0.30 \times 5.0 + 0.20 \times 2.0 - 0.30 \times 2.6}{0.20}$$

$$v = 5.6 \text{ m s}^{-1} \textit{ [1 mark]}$$

(ii) To determine whether the collision of the spheres is inelastic or perfectly elastic:

- Calculate the initial kinetic energy of the spheres

$$E_K = \tfrac{1}{2}mv^2$$

$$\text{initial } E_K = \tfrac{1}{2} \times 0.30 \times 5.0^2 + \tfrac{1}{2} \times 0.20 \times 2.0^2$$

$$\text{initial } E_K = 4.15 \text{ J}$$

- Calculate the final kinetic energy of the spheres

$$\text{final } E_K = \tfrac{1}{2} \times 0.30 \times 2.6^2 + \tfrac{1}{2} \times 0.20 \times 5.6^2$$

$$\text{final } E_K = 4.15 \text{ J } \textit{[1 mark]}$$

- Write a concluding statement

Since initial E_K = final E_K, the collision is elastic [1 mark]

1 mark for either value of initial E_K **OR** final E_K correct

1 mark for the values of initial E_K and final E_K both correct **AND** evidence that the collision is inelastic

Question 10 Hard [22 marks]

Part (a) [3 marks]

A stationary firework explodes into three different fragments that move in a horizontal plane. Fig. 1.1. shows the masses and velocities of the fragments A, B and C in terms of *M* and *v*.

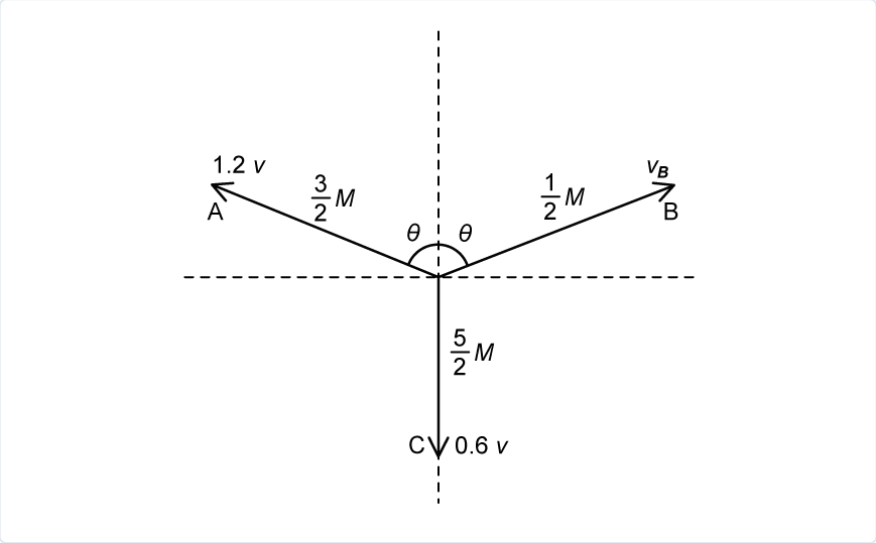


Fig. 1.1

Use the principle of conservation of momentum to determine the speed v_B of fragment B in terms of *v*.

Mark Scheme (3 marks):

Determine the speed of fragment B:

Resolve the momenta horizontally:

*Any **one** from:*

- Horizontal momentum of A: $p_A = \frac{3}{2}M \times 1.2v \sin\theta$ *[1 mark]*
- Horizontal momentum of B: $p_B = \frac{1}{2}M \times v_B \sin\theta$ *[1 mark]*
- Horizontal momentum of C: $p_C = 0$ *[1 mark]*

Apply conservation of momentum in the horizontal direction:

- Total momentum before = total momentum after
  - Total momentum before = 0
  - Total momentum after = $p_A - p_B$
- $p_A - p_B = 0$
- $p_A = p_B$
- $\frac{3}{2}M \times 1.2v \sin\theta = \frac{1}{2}M \times v_B \sin\theta$ *[1 mark]*

Simplify and calculate v_B:

- $v_B = 2 \times \frac{3}{2} \times 1.2v$
- $v_B = 3.6v$ *[1 mark]*

[Total: 3 marks]

Remember that the momenta are vectors and as such have a specific sign for their direction.

$+p_A$ *and* $-p_B$

Part (b) **[3 marks]**

Use the principle of conservation of momentum to determine the angle θ.

Mark Scheme (3 marks):

Determine the angle θ:

Resolve the momenta vertically:

*Any **one** from:*

- Vertical momentum of A: $p_A = \frac{3}{2}M \times 1.2v \cos \theta$ *[1 mark]*
- Vertical momentum of B: $p_B = \frac{1}{2}M \times 3.6v \cos \theta$ *[1 mark]*
- Vertical momentum of C: $p_C = \frac{5}{2}M \times 0.6v$ *[1 mark]*

Apply conservation of momentum in the vertical direction:

- Conservation of momentum: total momentum before = total momentum after
  - Total momentum before = 0
  - Total momentum after = $p_A + p_B - p_C$
- $p_A + p_B - p_C = 0$
- $p_A + p_B = p_C$
- $\frac{3}{2}M \times 1.2v \cos \theta + \frac{1}{2}M \times 3.6v \cos \theta = \frac{5}{2}M \times 0.6v$ *[1 mark]*

Simplify and calculate θ:

- $\frac{3}{2} \times 1.2 + \frac{1}{2} \times 3.6 \cos \theta = \frac{5}{2} \times 0.6$
- $\cos \theta = \frac{1.5}{3.6}$
- θ = 65.38 = 65° *[1 mark]*

[Total: 3 marks]

Part (c) **[4 marks]**

Another stationary firework explodes into four fragments which travel in different directions in a horizontal plane.

Fig. 1.2 shows the velocity and mass of each fragment.

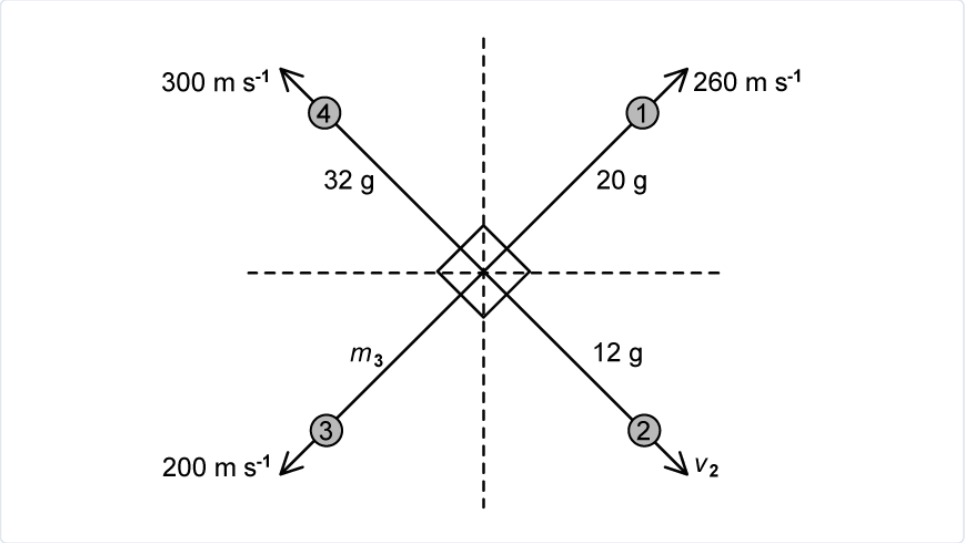


Fig. 1.2

Apply the principle of conservation of momentum to determine

(i) the velocity v_2

[2]

(ii) the mass m_3

[2]

Mark Scheme (4 marks):

List the known quantities:

- Fragment (1): mass m_1 = 20 g, velocity v_1 = 260 m s^{-1}
- Fragment (2): mass m_2 = 12 g, velocity = v_2
- Fragment (3): mass = m_3, velocity v_3 = 200 m s^{-1}
- Fragment (4): mass m_4 = 32 g, velocity v_4 = 300 m s^{-1}

(i) Determine the velocity of fragment (2):

- Momentum of (4): $p_4 = m_4 v_4 = 32 \times 300$
- Momentum of (2): $p_2 = m_2 v_2 = 12 \times v_2$

Apply conservation of momentum:

- Conservation of momentum (taking p_2 as the positive direction):
  $0 = p_2 - p_4 \quad \Rightarrow \quad p_2 = p_4$

- $12 \times v_2 = 32 \times 300$ *[1 mark]*
- $v_2 = \frac{32 \times 300}{12} = 800 \text{ m s}^{-1}$ *[1 mark]*

(ii) Determine the mass of fragment (3):

- Momentum of (1): $p_1 = m_1 v_1 = 20 \times 260$
- Momentum of (3): $p_3 = m_3 v_3 = m_3 \times 200$

Apply conservation of momentum:

- Conservation of momentum (taking p_1 as the positive direction):
  $0 = p_1 - p_3 \quad \Rightarrow \quad p_1 = p_3$
- $20 \times 260 = m_3 \times 200$ *[1 mark]*
- $m_3 = \frac{20 \times 260}{200} = 26 \text{ g}$ *[1 mark]*

[Total: 4 marks]

This kind of explosion question can look daunting at first, but taking some time to understand what is going on reveals the problem can be simplified by isolating the fragments and focusing on each line of action one at a time

Since the fragments all move at 90° to one another, you can consider the directions independently, similar to taking 'horizontal' and 'vertical' components in other vector problems. Once you realise this, then you can see that the fragments on the same line have ***equal momenta***

If you fancy a real mathsy challenge, have a go at part (e) of this question to see if you can truly get your head around how the momentum conservation laws can be applied to this sort of problem

Part (d) **[4 marks]**

Both fireworks have the same initial mass and release the same amount of kinetic energy.

Calculate

(i) the value of *M*

[1]

(ii) the magnitude of *v*.

[3]

Mark Scheme (4 marks):

List the known quantities:

- Fragment (A): mass = $\frac{3}{2}M$, velocity = $1.2 \quad v$
- Fragment (B): mass = $\frac{1}{2}M$, velocity = $3.6 \quad v$
- Fragment (C): mass = $\frac{5}{2}M$, velocity = $0.6 \quad v$
- Fragment (1): mass = 20 g, velocity = 260 m s^{-1}

- Fragment (2): mass = 12 g, velocity = 800 m s^{-1}
- Fragment (3): mass = 26 g, velocity = 200 m s^{-1}
- Fragment (4): mass = 32 g, velocity = 300 m s^{-1}

(i) Calculate the value of *M*:

- Sum of the masses (from **(b)**) = $\frac{3}{2}M + \frac{1}{2}M + \frac{5}{2}M = \frac{9}{2}M$
- Sum of the masses (from **(c)**) = 20 + 12 + 26 + 32 = 90 g
- Therefore, $\frac{9}{2}M = 90 \text{ g} \Rightarrow M = \frac{90}{4.5} = 20 \text{ g}$ *[1 mark]*

(ii) Calculate the magnitude of *v*:

Total kinetic energy of fragments A, B and C:

- $E_k = \frac{1}{2}Mv^2\frac{3}{2} \times 1.2^2 + \frac{1}{2} \times 3.6^2 + \frac{5}{2} \times 0.6^2$
- $E_k = 4.77Mv^2$ *[1 mark]*

Total kinetic energy of fragments 1, 2, 3 and 4:

- $E_k = \frac{1}{2}0.020 \times 260^2 + 0.012 \times 800^2 + 0.026 \times 200^2 + 0.032 \times 300^2$
- $E_k = 6476$ J *[1 mark]*

Determine the value of v:

- Total KE: $4.77Mv^2 = 6476$
- $v = \sqrt{\frac{6476}{4.77 \times 0.020}}$
- $v = 261 = 260$ m s^{-1} *[1 mark]*

[Total: 4 marks]

The key is in the question:

- *both fireworks have the **same initial mass***

This means the total mass of fragments A, B and C is equal to the total mass of fragments 1, 2, 3 and 4

- *both fireworks release the **same amount of kinetic energy***

This means the total KE of fragments A, B and C is equal to the total KE of fragments 1, 2, 3 and 4

To get the total KE of fragments A, B and C, start by writing the kinetic energy of each in terms of M and v and then simplify:

$$E_k = \frac{1}{2}\frac{3}{2}M \times 1.2v^2 + \frac{1}{2}\frac{1}{2}M \times 3.6v^2 + \frac{1}{2}\frac{5}{2}M \times 0.6v^2$$

$$E_k = \frac{1}{2}Mv^2\frac{3}{2} \times 1.2^2 + \frac{1}{2} \times 3.6^2 + \frac{5}{2} \times 0.6^2$$

$$E_k = \frac{1}{2}Mv^2 \times 9.54 = 4.77Mv^2$$

Part (e) [8 marks]

A firework similar to the one in **(c)** explodes into four fragments.

Fig. 1.3 shows the velocity and mass of each fragment.

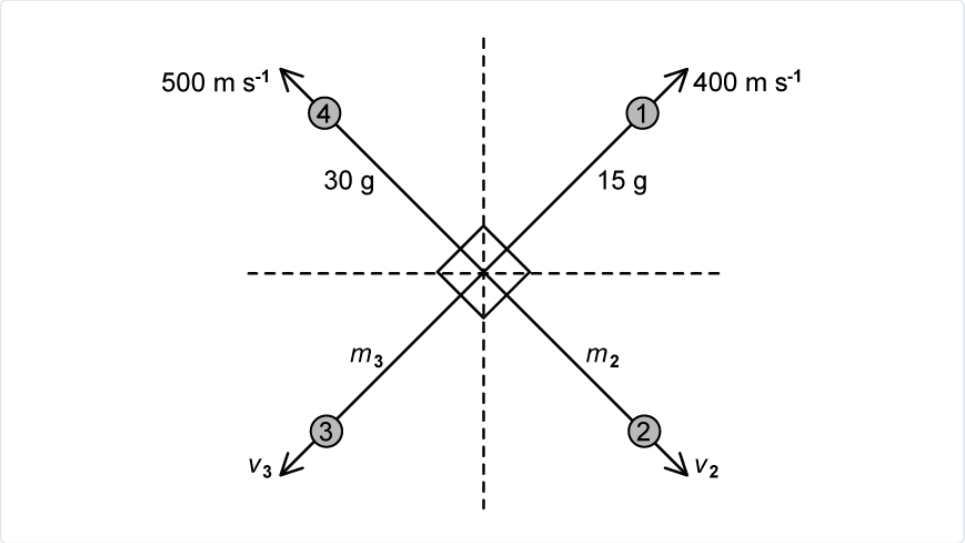


Fig. 1.3

(i) By applying the principle of conservation of momentum, show that

$$\frac{m_1}{m_2} = \frac{v_3}{v_4} = \frac{3}{5}$$

$$\frac{m_3}{m_4} = \frac{v_1}{v_2} = \frac{2}{3}$$

[6]

(ii) Hence, determine the values of m_2, v_2, m_3 and v_3.

[2]

Mark Scheme (8 marks):

(i) Apply conservation of momentum to Fig. 1.3:

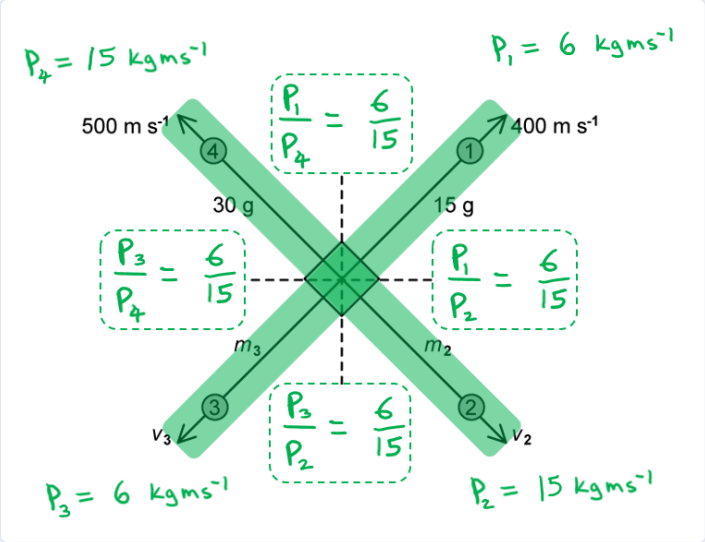
- Momentum of fragment 1: $p_1 = m_1 v_1 = 0.015 \times 400 = 6 \text{ kg m s}^{-1}$
- Momentum of fragment 4: $p_4 = m_4 v_4 = 0.03 \times 500 = 15 \text{ kg m s}^{-1}$

From conservation of momentum:

- Fragments on the same line have equal momenta, therefore: $p_1 = p_3 = 6 \text{ kg m s}^{-1}$ *[1 mark]*

  $p_2 = p_4 = 15 \text{ kg m s}^{-1}$ *[1 mark]*
- Sum of ↑ momenta = sum of ↓ momenta: $p_1 + p_4 = p_2 + p_3$
- Sum of → momenta = sum of ← momenta: $p_1 + p_2 = p_3 + p_4$
- So, $p_1 + p_4 = p_2 + p_3 = p_1 + p_2 = p_3 + p_4$ = constant

- Therefore, $\frac{p_1}{p_4} = \frac{p_2}{p_3} = \frac{p_1}{p_2} = \frac{p_3}{p_4}$ = constant
- Hence, $\frac{m_1 v_1}{m_4 v_4} = \frac{m_3 v_3}{m_2 v_2}$ **[Eq. 1]** *[1 mark]*
- Hence, $\frac{m_1 v_1}{m_2 v_2} = \frac{m_3 v_3}{m_4 v_4}$ **[Eq. 2]** *[1 mark]*



List the known ratios so far:

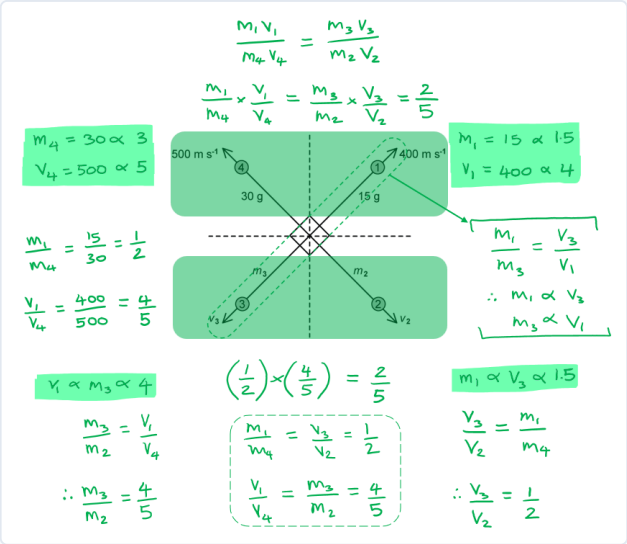
- From Fig. 1.2: $\frac{m_1}{m_4} = \frac{15}{30} = \frac{1}{2}$
- From Fig. 1.2: $\frac{v_1}{v_4} = \frac{400}{500} = \frac{4}{5}$
- Since $p_1 = p_3$, then $\frac{m_1}{m_3} = \frac{v_3}{v_1}$
- Since $p_2 = p_4$, then $\frac{m_2}{m_4} = \frac{v_4}{v_2}$
- From **[Eq. 1]** $\frac{m_1}{m_4} \times \frac{v_1}{v_4} = \frac{m_3}{m_2} \times \frac{v_3}{v_2} = \frac{6}{15} = \frac{2}{5}$
- From **[Eq. 2]** $\frac{m_1}{m_2} \times \frac{v_1}{v_2} = \frac{m_3}{m_4} \times \frac{v_3}{v_4} = \frac{6}{15} = \frac{2}{5}$

List the known proportional relationships so far:

- $m_1 \propto v_3 \propto 1.5$ (simplified from 15 g)
- $v_1 \propto m_3 \propto 4$
- $m_4 \propto v_2 \propto 3$
- $v_4 \propto m_2 \propto 5$ (simplified from 500 m s^{-1})

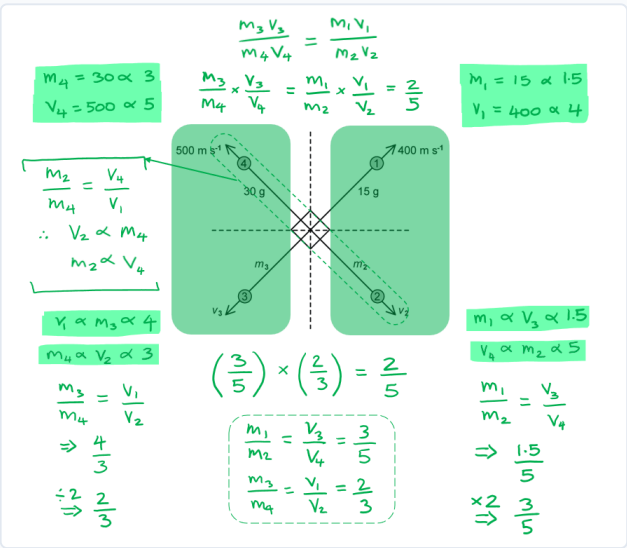
Use the proportional relationships to find the ratios in **[Eq. 1]**:

- $m_1 \propto v_3$ and $m_4 \propto v_2$ therefore $\frac{m_1}{m_4} = \frac{v_3}{v_2} = \frac{1.5}{3} = \frac{1}{2}$
- $v_1 \propto m_3$ and $m_2 \propto v_4$ therefore $\frac{m_3}{m_2} = \frac{v_1}{v_4} = \frac{4}{5}$



Use the proportional relationships to find the ratios in **[Eq. 2]**:

- $m_1 \propto v_3$ and $m_2 \propto v_4$ and from the question $\frac{m_1}{m_2} = \frac{v_3}{v_4} = \frac{3}{5}$: therefore $\frac{m_1}{m_2} = \frac{v_3}{v_4} = \frac{1.5}{5} \times 2 = \frac{3}{5}$ *[1 mark]*
- $m_3 \propto v_1$ and $m_4 \propto v_2$ and from the question $\frac{m_3}{m_4} = \frac{v_1}{v_2} = \frac{2}{3}$ therefore $\frac{m_3}{m_4} = \frac{v_1}{v_2} = \frac{4}{3} \div 2 = \frac{2}{3}$ *[1 mark]*



(ii) Hence, the missing values are:

- $\frac{m_1}{m_2} = \frac{3}{5} \Rightarrow m_2 = \frac{5}{3} \times 0.015 = 0.025\ 25\ \text{g}$

AND

- $\frac{v_3}{v_4} = \frac{3}{5} \Rightarrow v_3 = \frac{3}{5} \times 500 = 300\ \text{m s}^{-1}$ *[1 mark]*
- $\frac{m_3}{m_4} = \frac{2}{3} \Rightarrow m_3 = \frac{2}{3} \times 0.030 = 0.020\ 20\ \text{g}$

AND

- $\frac{v_1}{v_2} = \frac{2}{3} \Rightarrow v_2 = \frac{3}{2} \times 400 = 600\ \text{m s}^{-1}$ *[1 mark]*

> ***[Total: 8 marks]***

Question 11 Hard [18 marks]

Part (a) [4 marks]

A pellet of mass 2.7 g is fired into a stationary block of mass 540 g suspended from a rigid support as shown in Fig. 1.1.

The pellet becomes completely embedded in the block. The block can swing freely at the end of a light inextensible string of length 1.5 m measured from the pivot to the centre of the block.

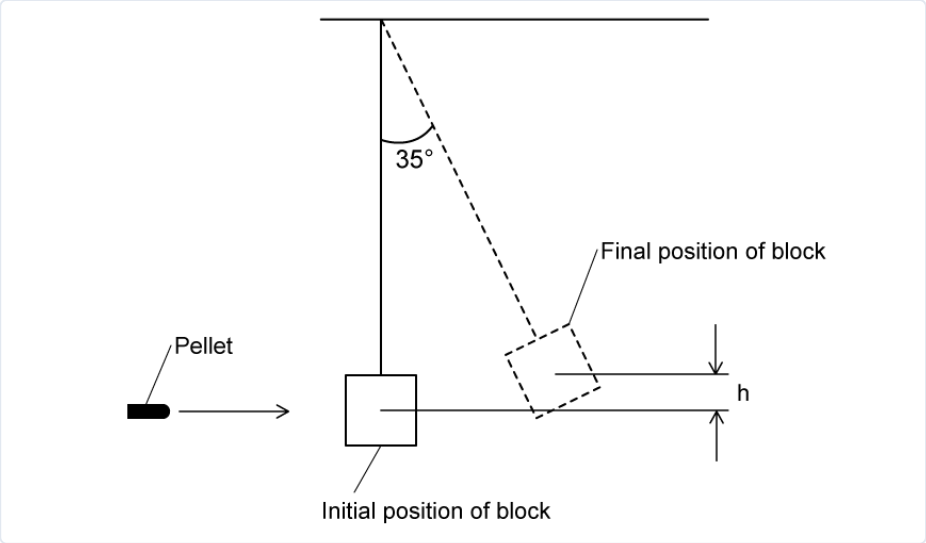


Fig. 1.1

The centre of mass of the block rises by a height h at an angle of 35° to the vertical.

Determine the velocity of the pellet and the block immediately after the pellet is embedded.

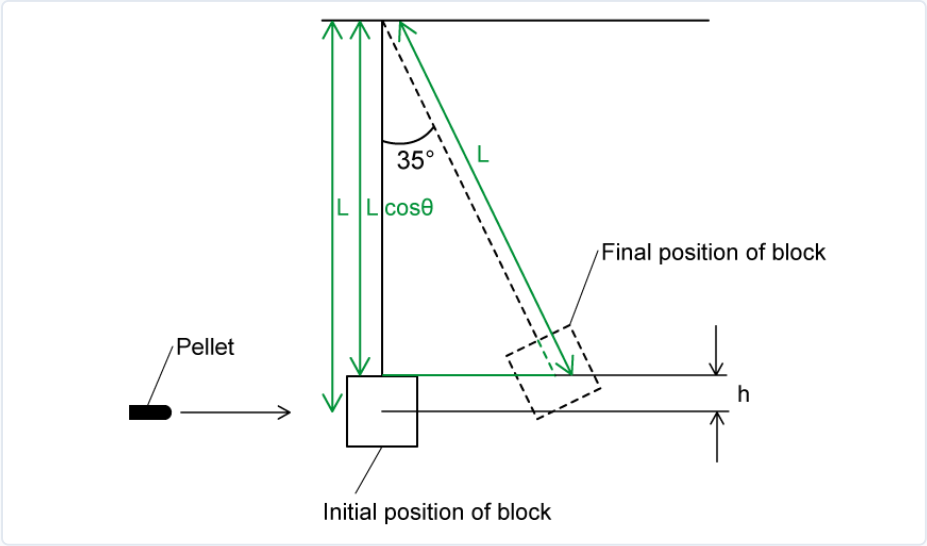
Mark Scheme (4 marks):

Determine the velocity of the pellet and the block immediately after the pellet is embedded:

List the known quantities:

- Mass of pellet = 2.7 g
- Mass of block = 540 g
- Combined mass, M = 540 + 2.7 = 542.7 g
- Angle between string and vertical, θ = 35°
- Length of string, L = 1.5 m

Calculate the change in height, h:



- From the diagram: $L = L\cos\theta + h \Rightarrow h = L - L\cos\theta$
- $h = 1.5 - 1.5\cos 35° = 0.271$ m *[1 mark]*

Calculate the initial speed of the block and the pellet:

- From conservation of energy: Initial KE = final GPE
- Initial KE = $\frac{1}{2}Mu^2$ **AND** final GPE = Mgh *[1 mark]*
- Equate and rearrange: $\frac{1}{2}Mu^2 = Mgh \Rightarrow u = \sqrt{2gh}$
- Substitute: $u = \sqrt{2 \times 9.81 \times 0.271}$ *[1 mark]*
- Initial speed of pellet and block: u = 2.306 = 2.3 m s^{-1} *[1 mark]*

[Total: 4 marks]

This type of question is known as a 'ballistic pendulum' and the marks are given for the method, as well as the final answer, so make sure you set out your working clearly!

*The 'initial kinetic energy' is treated as when the pellet first strikes and completely embeds itself in the block. Therefore, the mass is always the **sum** of the mass of the pellet and block.*

You can see that when you equate KE and GPE algebraically, the mass term cancels, so if you're confident doing this, it will definitely save you some time when doing the calculation step!

Part (b) [3 marks]

Calculate the velocity of the pellet just before it strikes the block.

Mark Scheme (3 marks):

Calculate the velocity of the pellet just before it strikes the block:

List the known quantities:

- Mass of the pellet, m_p = 2.7 g = 0.0027 kg

- Mass of the block, m_b = 540 g = 0.540 kg
- Combined mass, M = 542.7 g = 0.5427 kg
- Initial velocity of pellet and block, u = 2.306 m s^{-1} (from **(a)**)

Apply the principle of conservation of momentum:

- Total momentum lost by pellet = total momentum gained by pellet and block
- Total momentum lost by pellet = $m_p v_p$
- Total momentum gained by pellet and block = $m_p + m_b u = Mu$
- $m_p v_p = m_p + m_b u \Rightarrow m_p v_p = Mu$ *[1 mark]*
- $v_p = \frac{Mu}{m_p} = \frac{0.5427 \times 2.306}{0.0027}$ *[1 mark]*
- v_p = 463.506 = 464 m s^{-1} *[1 mark]*

[Total: 3 marks]

Part (c) **[6 marks]**

The block is replaced by a metal block of the same mass. The experiment is repeated with the wooden block and an identical pellet. The pellet rebounds after striking the block.

A student makes an assumption that the angle that the metal block makes with the vertical will be greater than 35° because the block doesn't have the additional mass of the pellet embedded within it.

Discuss the validity of the student's assumptions.

Mark Scheme (6 marks):

For the assumption:

The angle that the metal block makes with the vertical will be greater than 35° because the block doesn't have the additional mass of the pellet embedded within it

State whether the student's reasoning is correct or incorrect:

- The student is correct that the angle will increase; *[1 mark]*
- (However) the student's explanation / reasoning is incorrect; *[1 mark]*

Some valid comments about the student's assumption are:

*Any **two** pairs from:*

- The change in momentum of the metal block will be greater; *[1 mark]*
- (Because) a rebounding pellet causes a greater change in momentum; *[1 mark]*
- The speed / kinetic energy of the metal block will be greater (after the pellet has rebounded); *[1 mark]*
- (Because) the momentum / mass / speed of the pellet is still the same; *[1 mark]*

- (From conservation of energy) all the initial KE is transferred to the GPE of the block; *[1 mark]*
- (Hence) the height reached by the metal block will be greater (than the height reached by the original block); *[1 mark]*

[Total: 6 marks]

Part (d) **[5 marks]**

A popular demonstration of the conservation of momentum and conservation of energy is Newton's cradle. It features several identical polished steel balls hung in a straight line in contact with each other, as shown in Fig. 1.2.

If one ball is pulled back and allowed to strike the line, one ball is released from the other end whilst the rest are stationary. If two are pulled out, two are released on the other end and so forth.

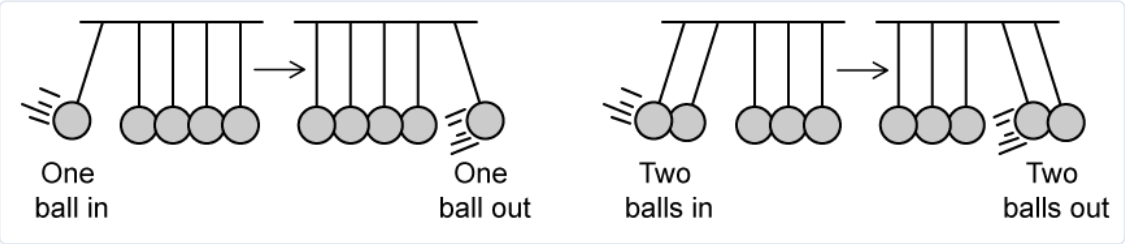


Fig. 1.2

Explain why swinging one ball from the left will not release two balls on the right.

Assume that the demonstration takes place in a vacuum.

Mark Scheme (5 marks):

Swinging one ball from the left will not release two balls on the right because:

- This would violate the conservation of energy / energy will not be conserved;

  OR

- It is an inelastic collision but should be elastic (in a vacuum); *[1 mark]*

Apply conservation of momentum:

- Conservation of momentum: total momentum before = total momentum after
- Total momentum before = *mu*
- Total momentum after = 2*mv*
- Conservation of momentum: $mu = 2mv \Rightarrow v = \frac{u}{2}$ *[1 mark]*
- Therefore, conservation of momentum is not violated if the final speed of the two balls is halved; *[1 mark]*

Apply conservation of energy:

- In a vacuum, there is no energy lost to the surroundings
- Hence, elastic collision: total kinetic energy before = total kinetic energy after
- Total kinetic energy before = $\frac{1}{2}mu^2$
- Total kinetic energy after = $\frac{1}{2}2m\frac{u}{2}^2 \quad = \quad \frac{1}{4}mu^2$ *[1 mark]*
- Therefore, conservation of energy is violated as half the kinetic energy is lost; *[1 mark]*

[Total: 5 marks]

Question 12 Hard [13 marks]

Part (a) [2 marks]

A body of mass M and moving with velocity V explodes into three equal fragments each of mass $\frac{M}{3}$, as shown in Fig. 1.1.

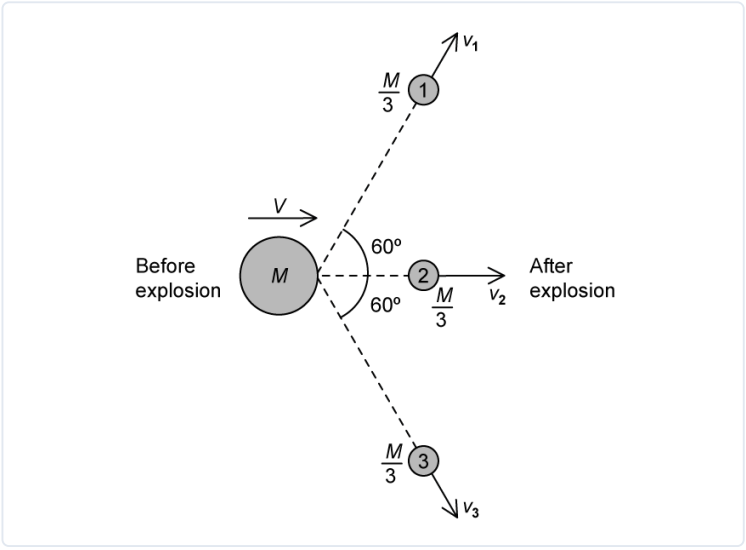


Fig. 1.1

Fragment 2 continues to move along the horizontal plane with velocity v_2, while fragments 1 and 3 move off at an angle of 60° to the horizontal with velocities v_1 and v_3 respectively, as shown.

By applying the principle of conservation of momentum in the vertical plane, show that $v_1 = v_3$

Mark Scheme (2 marks):

Show that $v_1 = v_3$ by considering momentum in the vertical plane:

- Vertical plane: total momentum before = 0
- Vertical plane: total momentum after = $\frac{M}{3} v_1 \sin 60° - \frac{M}{3} v_3 \sin 60°$ *[1 mark]*

Apply conservation of momentum:

- Conservation of momentum: total momentum before = total momentum after
- $0 = \frac{M}{3} v_1 \sin 60° - \frac{M}{3} v_3 \sin 60°$
- Simplify: $\frac{M}{3} v_1 \sin 60° = \frac{M}{3} v_3 \sin 60° \Rightarrow v_1 = v_3$ *[1 mark]*

[Total: 2 marks]

Part (b) [3 marks]

By applying conservation of momentum in the horizontal plane, show that $3V = v_1 + v_2$

Mark Scheme (3 marks):

Show that $3V = v_1 + v_2$ by considering momentum in the horizontal plane:

- Horizontal plane: total momentum before = MV
- Horizontal plane: total momentum after = $\frac{M}{3} v_1 \cos 60° + \frac{M}{3} v_2 + \frac{M}{3} v_3 \cos 60°$ *[1 mark]*

Apply conservation of momentum:

- Conservation of momentum: total momentum before = total momentum after
- $MV = \frac{M}{3} v_1 \cos 60° + \frac{M}{3} v_2 + \frac{M}{3} v_3 \cos 60°$
- Substitute $v_1 = v_3$ in: $MV = 2 \times \frac{M}{3} v_1 \cos 60° + \frac{M}{3} v_2$ *[1 mark]*
- Simplify: $V = 2 \times \frac{1}{3} v_1 \cos 60° + \frac{1}{3} v_2 \Rightarrow 3V = v_1 + v_2$ *[1 mark]*

[Total: 3 marks]

The full worked solution is shown below:

$$MV = \frac{M}{3} v_1 \cos 60° + \frac{M}{3} v_2 + \frac{M}{3} v_3 \cos 60°$$

$$V = 2 \times \frac{1}{3} v_1 \cos 60° + \frac{1}{3} v_2$$

$$3V = 2v_1 \cos 60° + v_2$$

$$\cos 60° = \frac{1}{2}$$

$$3V = 2 \times \frac{1}{2} v_1 + v_2$$

$$3V = v_1 + v_2$$

Part (c) **[4 marks]**

The ratio of the kinetic energy of fragment 1 to mass *M* is

$$\frac{kinetic\ \ energy\ \ of\ \ fragment\ \ 1}{kinetic\ \ energy\ \ of\ \ mass\ \ M} = \frac{4}{3}$$

Express velocities v_1 and v_2 in terms of the velocity V of mass M.

Mark Scheme (4 marks):

Express velocities v_1 and v_2 in terms of V:

- Kinetic energy of mass *M*: $KE_i = \frac{1}{2} MV^2$
- Kinetic energy of fragment 1: $KE_1 = \frac{1}{2} \frac{M}{3} {v_1}^2$ *[1 mark]*

Substitute the expressions into the ratio:

- $\frac{KE_1}{KE_i} = \frac{4}{3} \Rightarrow KE_1 = \frac{4}{3} KE_i$

- $\frac{1}{2}\frac{M}{3}{v_1}^2 = \frac{4}{3}\frac{1}{2}MV^2$ *[1 mark]*

Simplify to determine the velocity of fragment 1:

- $\frac{1}{3}{v_1}^2 = \frac{4}{3}V^2 \quad \Rightarrow \quad {v_1}^2 = 4V^2$
- Velocity of fragment 1: $v_1 = 2V$ *[1 mark]*

Determine the velocity of fragment 2:

- $3V = v_1 + v_2 \quad \Rightarrow \quad v_2 = 3V - 2V$
- Velocity of fragment 2: $v_2 = V$ *[1 mark]*

[Total: 4 marks]

Part (d) **[4 marks]**

(i) Determine the total kinetic energy transferred to the fragments.

[3]

(ii) Suggest why the total kinetic energy of the fragments is greater than the initial kinetic energy of the original mass.

[1]

Mark Scheme (4 marks):

(i) Determine the total kinetic energy transferred to the fragments:

- Kinetic energy of fragment 1: $KE_1 = \frac{4}{3}KE_i$
- Kinetic energy of fragment 3: $KE_3 = \frac{4}{3}KE_i$ (since $v_1 = v_3$)
- Where Kinetic energy of mass *M*: $KE_i = \frac{1}{2}MV^2$ (from part (c))

Determine the proportion of KE transferred to fragment 2:

- $\alpha KE_i = KE_2$ where α is the proportion of KE transferred to the fragment
- $\alpha\frac{1}{2}MV^2 = \frac{1}{2}\frac{M}{3}V^2$ (since $v_2 = V$)
- $\alpha\frac{1}{2}MV^2 = \frac{1}{3}\frac{1}{2}MV^2 \quad \Rightarrow \quad \alpha = \frac{1}{3}$
- Therefore, $KE_2 = \frac{1}{3}KE_i$ *[1 mark]*

Determine the total kinetic energy:

- Total initial KE = $KE_1 + KE_2 + KE_3$
- Total initial KE = $\frac{4}{3}KE_i + \frac{1}{3}KE_i + \frac{4}{3}KE_i$ *[1 mark]*
- Total initial KE = $3KE_i$ *[1 mark]*

(ii) The kinetic energy of the fragments is greater than the initial kinetic energy because:

- The total energy before the explosion is equal to the kinetic energy of mass *M* plus...

*Any **one** from:*

- Chemical energy (stored in mass *M*); *[1 mark]*
- Thermal energy (from the explosion); *[1 mark]*
- Internal energy (stored in mass *M*); *[1 mark]*
- Work done (from the explosion); *[1 mark]*

[Total: 4 marks]